

The Placental Response to Excess Maternal Glucocorticoid Exposure Differs Between the Male and Female Conceptus in Spiny Mice¹

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ABSTRACT

The placenta is the intermediary between the mother and fetus, and its primary role is to provide for the appropriate growth of the fetus. A suboptimal in utero environment has been shown to differentially affect the health of offspring, depending on their sex. Here we show that excess maternal glucocorticoids administered in midgestation (Day 20, 0.5 gestation in the spiny mouse) for 60 h, have persisting effects on the placenta that differ by fetal sex, placental region, and time after glucocorticoid exposure. Dexamethasone (DEX) exposure altered placental structure and mRNA expression from male and female fetuses both immediately (Day 23) and 2 wk posttreatment (Day 37). The immediate consequences (Day 23) of DEX were similar between males and females, with reductions in the expression of *IGF1*, *IGF1R*, and *SLC2A1* in the placenta. However, by Day 37, the transcriptional and structural response of the placenta was dependent on the sex of the fetus, with placentas of male fetuses having an increase in *GCM1* expression, a decrease in *SLC2A1* expression, and an increase in the amount of maternal blood sinusoids in the DEX-exposed placenta. Female placentas, on the other hand, showed increased *SLC2A1* and *MAP2K1* expression and a decrease in the amount of maternal blood sinusoids in response to DEX exposure. We have shown that the effect of a brief glucocorticoid exposure at midgestation has persisting effects on the placenta, and this is likely to have ongoing and dynamic effects on fetal development that differ for a male and female fetus.

developmental origins of health and disease, gene expression, sex, stress, trophoblast

INTRODUCTION

Maternal stress during pregnancy can have deleterious and long-lasting effects on the fetus and newborn, and this is often attributed to increases in the concentration of cortisol in the maternal circulation [1]. The fetus is somewhat protected from changes in maternal cortisol levels by the presence of hydroxysteroid (11-beta) dehydrogenase 2 (HSD11B2) in the

placenta, which converts cortisol to the less active cortisone [2]. Nevertheless, it is known that events that increase glucocorticoid secretion during gestation, such as maternal physical or psychological stress, or maternal under-nutrition, can have significant effects on fetal growth and development, which, ultimately, affects the health and wellbeing of the offspring [3–6]. In addition to a number of experimental studies demonstrating the harmful effects of excess glucocorticoids on fetal and postnatal development [3, 5, 7, 8], a striking example comes from the 2001 terrorist attacks on the World Trade Center in New York. For pregnant women in the New York area at that time, there was a higher incidence of low birth weight babies, with the greatest proportion arising from mothers who were in the first two trimesters of their pregnancy at the time of the attack [9]. There was also an increased incidence of male fetal death in New York after September 2001 [10, 11].

In animal models, maternal glucocorticoids administered for prolonged periods during gestation generally result in low birth weight offspring [3, 12]. Short periods of high maternal glucocorticoids given during early to midgestation tend to impact on organ development rather than birth weight; for example, nephron number in the kidney is reduced, often resulting in increased blood pressure in the offspring [13–15]. We have reported that this reduction in nephron number is caused, at least in part, by a reduction in branching morphogenesis during ureteric development, so that fewer glomeruli are formed during fetal development [15]. More recently, in our model of excess maternal glucocorticoids at midgestation, we have found evidence of altered steroidogenic function in the adrenal cortex of adult male and female spiny mouse offspring [16].

While much is known about the effects of high glucocorticoids on the fetus itself, less is known about effects that may arise directly in the placenta. Many studies have reported a decrease in placental weight and/or an increase in fetal to placental weight ratio in studies focused mainly on fetal outcomes [17–20]. A comprehensive study in mice by Baisden et al. reported that multiple injections of dexamethasone (DEX) in the last quarter of gestation resulted in gross morphological and microscopic changes, and altered gene expression in the placenta [21]. Other studies using data obtained from animals, humans, and placental cell lines have reported alterations in placental transport of proteins [22, 23], expression of vasculogenic [24] and apoptosis-related genes [25, 26], and decreased placental steroidogenesis [27] in response to high glucocorticoid exposure.

A notable omission from these studies is an account of the changes in relation to the sex of the fetus. The aim of the current study was to determine whether excess maternal glucocorticoids alter the structure and transcriptional expres-

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sion of selected genes, related to those processes above, in placentas obtained from male and female fetuses. Specifically, we investigated whether the consequences for the placenta of excess maternal glucocorticoids would differ depending on the sex of the fetus, the region of the placenta examined, and/or the time elapsed after glucocorticoid exposure. In order to undertake this study, we used the spiny mouse, a rodent species in which the natural circulating glucocorticoid is cortisol [28].

MATERIALS AND METHODS

Animals

All experiments were approved in advance by the Monash University Animal Ethics Committee and were conducted in accordance with the Australian Code of Practice for the Care and Use of Animals for Scientific Purposes. The spiny mice used in this study were obtained from our own colony and bred as previously described [29]. The spiny mouse is a rodent species with a relatively long gestation (39 days), small litter size (1–5, normally 2–3), and precocial young; for example, when born, the offspring are furred, and have open eyes and sophisticated locomotor capabilities, and organogenesis of the major organs (liver [30], lung [31], brain [32], and kidney [33]) is largely complete by the time of birth.

DEX Administration

DEX is a synthetic glucocorticoid that binds preferentially to the glucocorticoid receptor, and not the mineralocorticoid receptor [34], and therefore mimics the effects of endogenous glucocorticoids acting through the glucocorticoid receptor. As previously described [15], pregnant dams at Day 20 (d20) of gestation were exposed to DEX (125 µg/kg in 75 µl) or saline (75 µl) infused for 60 h from a subcutaneously implanted osmotic pump (Model 1003D; Alzet). On d23, at the end of DEX or saline administration, the dams were allocated to one of two groups where fetal and placental tissues were collected immediately, or the osmotic pump was removed under brief anesthesia, and pregnancy allowed to continue until d37, when fetal and placental tissues were then collected.

Tissue Collection and Preparation

At d23 or d37 the dams were killed by cervical dislocation, the uterus rapidly excised, and the placentas and fetuses removed and weighed. Fetuses were sexed by visualization of gonads at d37, and, at d23, a piece of tail was frozen for later PCR of the sex-determining region (*Sry*) gene, as previously described [15]. All placentas were divided in half across the largest diameter. One-half was then separated into labyrinth and junctional zone regions by careful dissection, and the samples were snap frozen in liquid nitrogen. The other half of the placenta was fixed in 10% buffered formalin for 48 h. Tissues frozen in liquid nitrogen were subjected to a total RNA extraction technique using the RNeasy Mini Kit (Qiagen). Samples (1 µg RNA) were then reverse transcribed to form cDNA (Reverse Transcription System; Promega) and stored at –20°C until later use. Fixed placentas were dehydrated in a series of alcohols and then embedded in paraffin wax. Microtome sections (5-µm thick) were stained with an adapted Masson trichrome stain.

Quantitative Real-Time PCR Analysis

A TAQman multiplex assay was used to determine relative levels of mRNA expression for B-cell CLL/lymphoma 2 (*Bcl2*), while, for glial cell missing homolog 1 (*GCM1*), mitogen-activated protein kinase kinase 1 (*MAP2K1*), solute carrier family 2 (facilitated glucose transporter), member 1 (*SLC2A1*), BCL2-associated X protein (*Bax*), nuclear receptor subfamily 3, group C, member 1 (glucocorticoid receptor, *NR3C1*), insulin-like growth factor 1 receptor (*IGF1R*), and insulin-like growth factor 1 (*IGF1*), SYBR green PCR master mix was used, per the manufacturer's instructions (Applied Biosystems). The primer and probe sequences are described in Table 1. Primer design was performed as previously described [35]. Briefly, for each gene of interest, the cDNA sequences in *Mus musculus*, *Rattus norvegicus*, and *Homo sapiens* were aligned, and primers were designed based on identified regions of high homology. Optimizing PCR assays were conducted in order to test the specificity of the primer sets, which were found to amplify products of the expected sizes (data not shown). PCR products were then sequenced and confirmed to be the gene of interest using the Basic Local Alignment Search Tool (BLAST; <http://blast.ncbi.nlm.nih.gov>; data not shown). Quantitative real-

time PCR (qPCR) cycling conditions for all genes consisted of an initial denaturation step of 95°C for 10 min, followed by 40 cycles consisting of 95°C for 15 sec and 60°C for 1 min. A dissociation curve was also performed at the end of each run. Samples were run in triplicate for each gene of interest, and, where differences were observed between treatment groups, the qPCR was repeated.

Data obtained from qPCR were analyzed as previously described [35]. Briefly, the relative expression within the sample (ΔC_T) was calculated by subtracting the mean cycle threshold (C_T) for the housekeeping gene (RNA, 18S ribosomal 1 [*RN18S1*]) from the mean C_T value for the gene of interest. This number was then inserted into the formula, $2^{-\Delta C_T}$, to give a final arbitrary expression value. This expression value ($2^{-\Delta C_T}$) for each group was divided by the mean $2^{-\Delta C_T}$ of the saline male group to give a final value of expression, relative to the mean value of that obtained for the saline male group, for the gene of interest.

Tissue Analysis

Placental sections were placed into a microfiche reader (RF3A minicopy reader; Fuji, Japan) with a stereologic test grid (0.5 × 0.5 cm) and examined at 28.25× magnification. Points on the grid overlying placental tissue were designated as either labyrinth or junctional zone, based on gross morphological appearance. Within the junctional zone, regions were identified as either maternal blood sinusoids or trophoblast. Trophoblast populations within the junctional zone include the spongiotrophoblast, glycogen cells, and giant cells. At least three sections from each placenta were analyzed. Point counting continued until the analysis of an extra section did not change the mean value for each parameter.

Data Analysis and Statistics

All results are represented as mean ± SEM. Histological and fetal/placental weight parameters represent data from all dams from each treatment group at both time points; fetus number ranges from 6 to 13 of each sex, treatment (DEX or saline), and time point (d23 or d37). Quantitative real-time PCR data represent four or five fetuses of each sex, in both treatment groups, at both time points, and all were from different litters. Data generated by both histological and qPCR data were analyzed using a two-way ANOVA for treatment (P_{TREAT}) and sex (P_{SEX}). Where animals from the same litter were analyzed in the same group (histological analysis and fetal/placental weight parameters only), the mother was added as a variable to account for any variation due to maternal effects.

RESULTS

Sexually Dimorphic Observations in the Non-DEX-Treated Group

At d23 of gestation, female fetuses in the control (saline-treated) group were smaller than male fetuses ($P_{\text{SEX}} < 0.05$; Fig. 1A), and expression of the *GCM1* ($P_{\text{SEX}} < 0.05$; Table 2) and *IGF1R* ($P_{\text{SEX}} < 0.05$; Table 2) genes was higher in the female placenta compared to the male placenta. At d37, the junctional zone and the amount of trophoblast within the junctional zone were significantly greater in the male placenta compared to the female placenta (Fig. 2, D and G), whereas the labyrinth:junctional zone ratio was significantly greater in females compared to males (Fig. 2E). The ratio of *BAX:BCL2* expression at d37 in the junctional zone was higher in male placentas than female placentas ($P_{\text{SEX}} < 0.05$; Table 2).

Immediate and Long-Term Effects of DEX on Gross Placental Morphology and Fetal Weight

There was no significant effect of DEX exposure on fetal or placental weights at d23 or d37 (Fig. 1, A and B), although the fetoplacental weight ratio at d37 was significantly decreased by the DEX treatment ($P_{\text{TREAT}} < 0.01$; Fig. 1C). Histological analysis revealed that DEX exposure had increased the amount of junctional zone within the placenta of both male and female fetuses at d23 compared to the saline-treated controls ($P_{\text{TREAT}} < 0.05$; Fig. 2D), but not at d37. The amount of labyrinth

TABLE 1. Primer sequences for genes of interest.

Gene of interest	Species*	Forward primer sequence (5'–3')	Reverse primer sequence (5'–3')
<i>RN18S1</i>	<i>Mus musculus</i> , <i>Rattus norvegicus</i> , <i>Homo sapiens</i>	ACA CGG ACA GGA TTG ACA GA	CAA CTA GCT CCA CCA ACT AA
<i>Bax</i>	<i>Mus musculus</i>	TTG CTG ATG GCA ACT TCA ACT G	CTT TAG TGC ACA GGG CCT TGA G
<i>Bcl2</i>	<i>Mus musculus</i>	Mm00437783_m1 (Applied Biosystems)	
<i>GCM1</i>	<i>Mus musculus</i> , <i>Rattus norvegicus</i> , <i>Homo sapiens</i>	GCA AGA TTT ACC TGA GAC CC	AAA GAT GAA GCG TCC GTC GT
<i>SLC2A1</i>	<i>Mus musculus</i> , <i>Rattus norvegicus</i> , <i>Homo sapiens</i>	CCC CGC TTC CTG CTC ATC AA	GCC GAC CCT CTT CTT TCA TC
<i>NR3C1</i>	<i>Ovis aries</i>	ACT GCC CCA AGT GAA AAC AGA	GCC CAG TTT CTC CTG CTT AAT TAC
<i>IGF1</i>	<i>Mus musculus</i> , <i>Rattus norvegicus</i> , <i>Homo sapiens</i>	CTC CAG CCT CCT CAG ATC AC	TTG CTC GCC TTC AGC AGC TC
<i>IGF1R</i>	<i>Mus musculus</i> , <i>Rattus norvegicus</i> , <i>Homo sapiens</i>	GCT TGT TCT CCT CGC TGT AG	CTG GAC AAG CCG GAT AAC TG
<i>MAP2K11</i>	<i>Mus musculus</i> , <i>Rattus norvegicus</i> , <i>Homo sapiens</i>	AAG CAG AAG GTG GGA GAA CT	TTC AAA TCT GCT CTC TCT GC
<i>Sry</i>	<i>Mus musculus</i>	CAG CTA ACA CTG ATC TTT TC	TTA CTG AGC CAG AAT CAT AG

* Where multiple species are listed, gene sequences of these species were aligned and consensus regions were used to design primers.

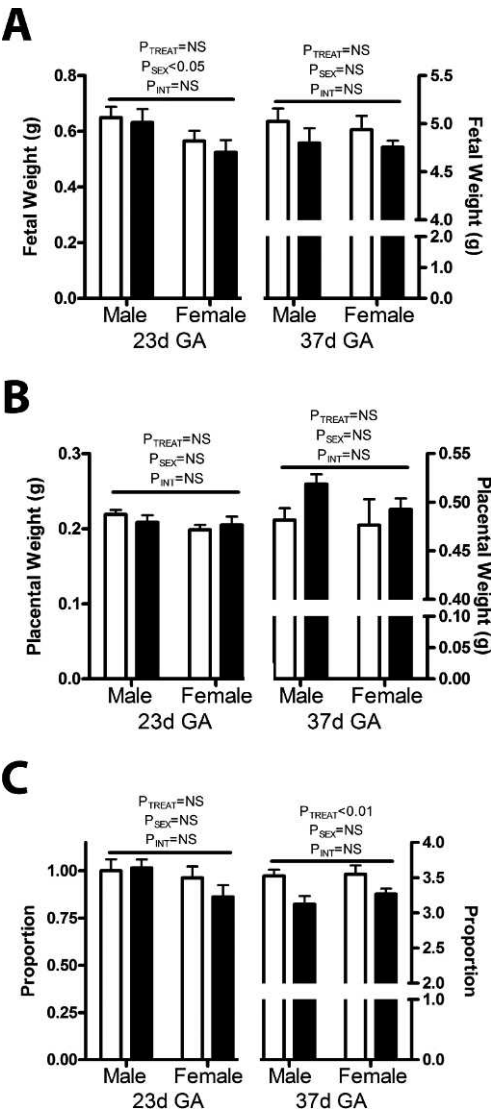


FIG. 1. Effects of maternal DEX exposure on fetal and placental weight. Weights of male and female fetuses (A), placentas (B), and the resultant fetal:placental weight ratio (C) immediately following maternal DEX exposure (closed bars; 23 dGA) and 2 wk later (37 dGA) compared to saline-treated controls (open bars). Graphs show mean values $\pm$ SEM. Two-way ANOVA for treatment (P_{TREAT}) and fetal sex (P_{SEX}); P_{INT} refers to an interaction effect between the treatment and sex effects. dGA, days gestation.

tissue in the placenta tended to be greater immediately after DEX exposure at d23 compared to controls ($P_{TREAT} = 0.06$; Fig. 2C); however, this difference was not present at d37. The labyrinth:junctional zone ratio was significantly decreased after DEX treatment ($P_{TREAT} < 0.05$; Fig. 2E) at d23, but not at d37. Within the junctional zone, the amount of trophoblast was increased in DEX-exposed placentas compared to controls ($P_{TREAT} < 0.05$; Fig. 2G) at d23, although the amount of the maternal blood sinusoids itself, and the sinusoid:trophoblast ratio, was not significantly altered (Fig. 2, F and H). However, at d37 the amount of the maternal blood sinusoid was increased significantly (+45%) in DEX-exposed male placentas, whereas, in female placentas, the maternal blood sinusoid was significantly decreased (–17%) by DEX treatment (Fig. 2F), resulting in a 22% increase in sinusoid:trophoblast ratio in males and a 16% decrease in sinusoid:trophoblast ratio in females ($P_{INT} < 0.001$; Fig. 2H) compared to saline-treated controls.

Effect of DEX on Gene Expression in the Labyrinth

Glucocorticoid receptor. Immediately following DEX exposure (d23), expression of the glucocorticoid receptor (*NR3C1*) within the labyrinth was significantly reduced in the placentas of both male and female fetuses ($P_{TREAT} < 0.01$; Table 2), but this treatment effect was not present 2 wk later, at d37 (Table 2).

Genes associated with nutrient delivery and fetal growth. The expression of *IGF1* and *IGF1R* were significantly decreased in both male and female placentas by DEX at both d23 and d37 (Table 2). Glucose transporter (*SCL2A1*) expression was reduced in both male and female placentas immediately after DEX exposure on d23 ($P_{TREAT} < 0.05$; Table 2), and this reduction was still present in the male placenta only (–53%) at d37 ($P_{INT} < 0.05$; Table 2).

Genes associated with patterning and proliferation. The expression of *GCM1* was not affected by DEX treatment at d23. However, at d37 *GCM1* expression was significantly reduced (–51%) in placentas of male fetuses, whereas it was increased (+68%) in placentas of female fetuses ($P_{INT} < 0.05$; Table 2). *MAP2K1* expression was significantly up-regulated at both d23 and d37 by DEX treatment in both males and females ($P_{TREAT} < 0.05$; Table 2). The ratio of *BAX:BCL2* expression in the labyrinth was decreased immediately after DEX exposure ($P_{TREAT} < 0.01$; Table 2) compared to saline-treated

TABLE 2. qPCR results for genes of interest.

Region	Gestational Day 23				Gestational Day 37			
	Male		Female		Male		Female	
	Saline (n = 5)	DEX (n = 5)	Saline (n = 5)	DEX (n = 4)	Saline (n = 5)	DEX (n = 4)	Saline (n = 4)	DEX (n = 5)
Labyrinth <i>NR3C1</i>	1.000 ± 0.181	0.510 ± 0.068	1.637 ± 0.334	0.505 ± 0.141	1.000 ± 1.032	0.729 ± 0.059	0.992 ± 0.084	1.069 ± 0.100
								$P_{TREAT} < 0.01$ $P_{SEX} = NS$ $P_{INT} = NS$
								$P_{TREAT} = NS$ $P_{SEX} = NS$ $P_{INT} = NS$
<i>IGF1</i>	1.000 ± 0.267	0.507 ± 0.050	1.257 ± 0.257	0.720 ± 0.162	1.000 ± 0.270	0.430 ± 0.089	0.770 ± 0.200	0.494 ± 0.158
								$P_{TREAT} < 0.05$ $P_{SEX} = NS$ $P_{INT} = NS$
								$P_{TREAT} = NS$ $P_{SEX} = NS$ $P_{INT} = NS$
<i>IGF1R</i>	1.000 ± 0.211	0.584 ± 0.054	1.777 ± 0.279	0.661 ± 0.143	1.000 ± 0.169	0.447 ± 0.075	0.757 ± 0.215	0.665 ± 0.106
								$P_{TREAT} < 0.01$ $P_{SEX} < 0.05$ $P_{INT} = NS$
								$P_{TREAT} = NS$ $P_{SEX} = NS$ $P_{INT} = NS$
<i>SLC2A1</i>	1.000 ± 0.175	0.631 ± 0.089	2.186 ± 0.690	0.691 ± 0.163	1.000 ± 0.140	0.472 ± 0.078	0.659 ± 0.122	0.680 ± 0.041
								$P_{TREAT} < 0.05$ $P_{SEX} = NS$ $P_{INT} = NS$
								$P_{TREAT} = NS$ $P_{SEX} = NS$ $P_{INT} = NS$
<i>GCM1</i>	1.000 ± 0.258	0.930 ± 0.131	2.191 ± 0.351	1.299 ± 0.332	1.000 ± 0.103	0.458 ± 0.102	0.548 ± 0.061	0.918 ± 0.203
								$P_{TREAT} = NS$ $P_{SEX} < 0.05$ $P_{INT} = NS$
								$P_{TREAT} = NS$ $P_{SEX} = NS$ $P_{INT} = NS$
<i>MAP2K1</i>	1.000 ± 0.534	2.122 ± 0.561	0.695 ± 0.534	4.043 ± 2.293	1.000 ± 0.211	0.870 ± 0.234	0.736 ± 0.238	0.832 ± 0.217
								$P_{TREAT} < 0.05$ $P_{SEX} = NS$ $P_{INT} = NS$
								$P_{TREAT} = NS$ $P_{SEX} = NS$ $P_{INT} = NS$
<i>BAX:BCL2</i>	1.000 ± 0.223	0.326 ± 0.072	1.655 ± 0.404	0.354 ± 0.090	1.000 ± 0.139	0.305 ± 0.088	1.012 ± 0.175	0.512 ± 0.213
								$P_{TREAT} < 0.01$ $P_{SEX} = NS$ $P_{INT} = NS$
								$P_{TREAT} = NS$ $P_{SEX} = NS$ $P_{INT} = NS$
Junctional zone <i>NR3C1</i>	1.000 ± 0.182	0.780 ± 0.194	1.010 ± 0.215	1.096 ± 0.382	1.000 ± 0.085	0.729 ± 0.147	0.882 ± 0.126	0.699 ± 0.118
								$P_{TREAT} = NS$ $P_{SEX} = NS$ $P_{INT} = NS$
								$P_{TREAT} = NS$ $P_{SEX} = NS$ $P_{INT} = NS$
<i>SLC2A1</i>	1.000 ± 0.156	1.442 ± 0.386	1.540 ± 0.319	0.443 ± 0.089	1.000 ± 0.146	1.038 ± 0.144	0.133 ± 0.014	1.453 ± 0.386
								$P_{TREAT} < 0.05$ $P_{SEX} = NS$ $P_{INT} = NS$
								$P_{TREAT} = NS$ $P_{SEX} = NS$ $P_{INT} = NS$
<i>MAP2K1</i>	1.000 ± 0.341	8.177 ± 4.158	1.005 ± 0.618	9.521 ± 1.295	1.000 ± 0.299	0.673 ± 0.124	0.650 ± 0.249	2.794 ± 1.047
								$P_{TREAT} < 0.01$ $P_{SEX} = NS$ $P_{INT} = NS$
								$P_{TREAT} = NS$ $P_{SEX} = NS$ $P_{INT} = NS$
<i>BAX:BCL2</i>	1.000 ± 0.263	0.739 ± 0.230	1.130 ± 0.105	0.248 ± 0.070	1.000 ± 0.087	1.526 ± 0.269	0.592 ± 0.088	0.830 ± 0.207
								$P_{TREAT} < 0.01$ $P_{SEX} = NS$ $P_{INT} = NS$
								$P_{TREAT} = NS$ $P_{SEX} < 0.05$ $P_{INT} = NS$

* All data are expressed relative to the male saline group at each gestational age. TREAT, treatment; NS, not significant; P_{INT} refers to an interaction effect between the treatment and sex effects.

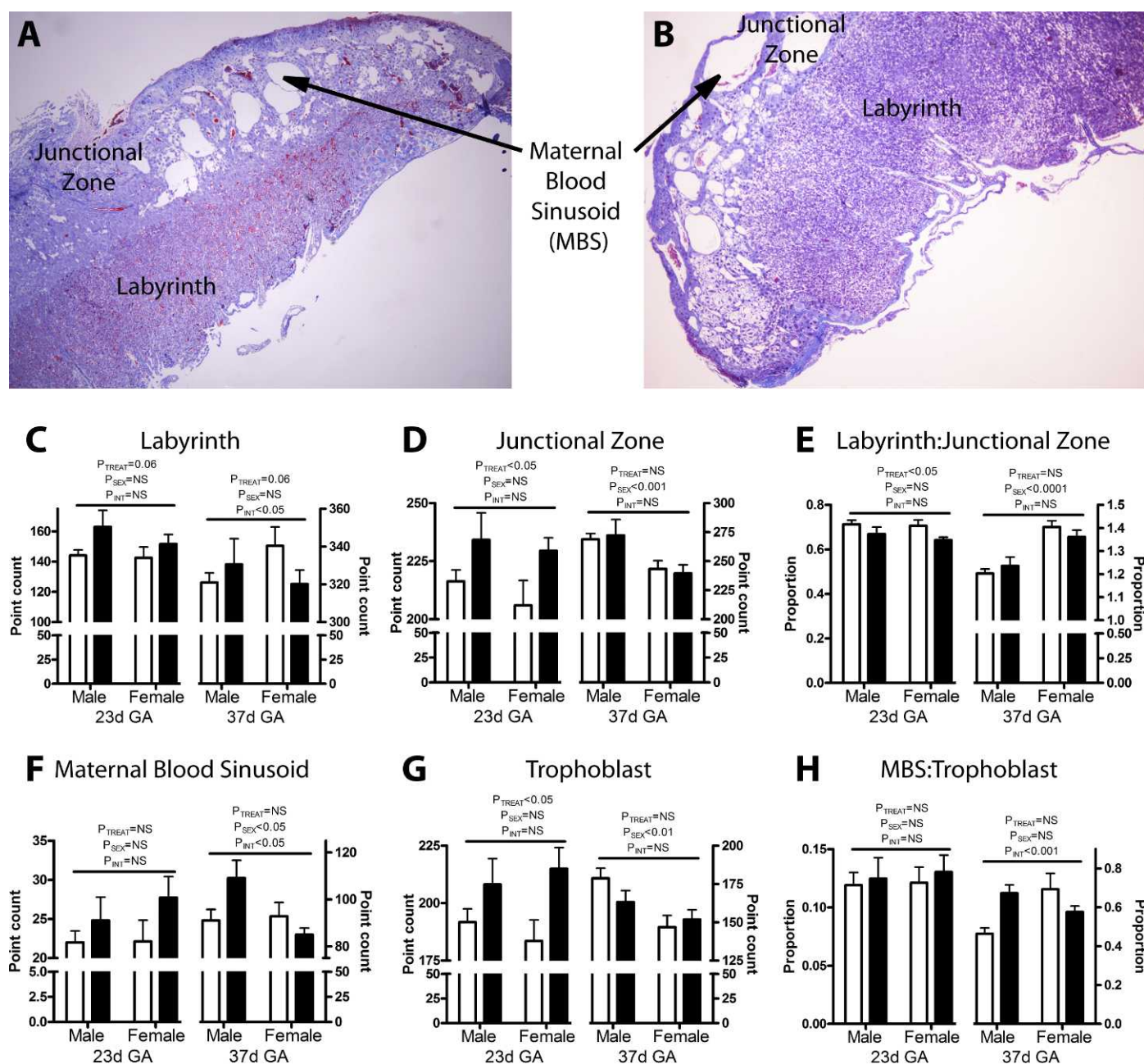


FIG. 2. Effects of maternal DEX exposure on placental structure. Histological micrographs of whole transverse sections of control 23dGA (A) and 37dGA (B) placentas stained with Masson trichrome, highlighting the morphological structures analyzed. Original magnification $\times 100$. Amount of labyrinth (C), junctional zone (D), and the labyrinth:junctional zone ratio (E) in the spiny mouse placenta immediately following maternal DEX exposure (closed bars; 23 dGA) and 2 wk later (37 dGA) compared to saline-treated controls (open bars). Within the junctional zone, the amount of maternal blood sinusoid (F), and trophoblast (G), along with the resulting maternal blood sinusoid:trophoblast ratio (H) immediately following maternal DEX exposure and 2 wk later (37 dGA) compared to saline-treated controls. Graphs show mean values $\pm$ SEM. Two-way ANOVA for treatment (P_{TREAT}) and fetal sex (P_{SEX}); P_{INT} refers to an interaction effect between the treatment and sex effects. dGA, days gestation.

controls, and this change, which was not different between the sexes, persisted to d37 ($P_{\text{TREAT}} < 0.01$; Table 2).

Effect of DEX on Gene Expression in the Junctional Zone

Glucocorticoid receptor. In contrast to the labyrinth, the expression of the glucocorticoid receptor (*NR3C1*) in the junctional zone was not affected by the DEX treatment (Table 2).

Genes associated with nutrient delivery and fetal growth. Immediately following cessation of the DEX treatment at d23, expression of the glucose transport (*SLC2A1*) was

increased (+44%) in the junctional zone of DEX-exposed male placentas, but decreased (−71%) in that of the female placenta ($P_{\text{TREAT}} < 0.05$, $P_{\text{INT}} < 0.05$; Table 2). However, at d37, *SLC2A1* expression was increased in the junctional zone of both male and female placentas, but the increase was significantly greater in that of female fetuses ($P_{\text{TREAT}} < 0.05$, $P_{\text{INT}} < 0.05$; Table 2).

Genes associated with patterning and proliferation. *MAP2K1* expression was significantly increased in both sexes immediately after the end of the DEX treatment (i.e., d23; $P_{\text{TREAT}} < 0.01$; Table 2), but this change did not persist until

term. Placentas of female fetuses tended to show increased *MAP2K1* expression in the junctional zone at d37, whereas placentas of males tended to show a small decrease in *MAP2K1* expression at d37 ($P_{\text{INT}} = 0.07$; Table 2). In both sexes, the ratio of *BAX:BCL2* expression was significantly reduced in the junctional zone following DEX treatment at d23 ($P_{\text{TREAT}} < 0.01$; Table 2), but, in contrast, it was significantly increased at d37 ($P_{\text{TREAT}} < 0.05$; Table 2).

DISCUSSION

This study has revealed that a short exposure to maternally administered DEX at midpregnancy has both immediate and long-term effects on placental structure and gene transcription. We also show that the placentas of male and female fetuses respond differently to this glucocorticoid exposure. A major advantage of the prolonged gestation in the spiny mouse is that we are able to observe long-term changes that occurred in the placenta, well after the DEX exposure had ceased (2 wk).

Sexually Dimorphic Observations in the Non-DEX-Treated Group

Male spiny mouse fetuses are larger than females at midgestation, suggesting that they, like the male conceptus of other species [36], grow faster than the female in early pregnancy. This sex difference can also be observed in the placenta, where the junctional zone, in terms of both maternal blood sinusoids and trophoblast, was greater in the male placenta compared to the female placenta near term. This suggests that the male is able to affect the development of the maternal vascular compartment of the placenta. In contrast, the placenta of the female fetus, in which the labyrinth:junctional zone ratio is greater than that of the male near term, may be more efficient in terms of extracting and transporting nutrients from the maternal circulation, but further studies of placental function will be necessary to confirm this. However, consistent with such speculation is the observation that *IGFIR* expression was increased in the labyrinth of female placentas, which suggests that the female placenta might be more sensitive to bioavailable IGFI early in gestation. This could promote fetal growth without the need to directly affect the delivery of nutrients to the placenta, which contrasts with our observations of male placental development.

Effect of DEX on Fetal Growth, Placental Structure, and Gene Expression

The reduction in glucocorticoid receptor gene expression immediately after exposure to DEX indicates that the dose given was biologically active and reached the placenta. Importantly, we did not observe any sex-specific differences in the expression of the glucocorticoid receptor gene, consistent with other studies [37], suggesting that this is unlikely to be the source of our observed sex differences.

We observed a number of gene changes in the placenta in response to DEX exposure, and, importantly, differences between the male and female placenta. *SLC2A1* is the predominant glucose transporter gene isoform in the placenta [38], and our data indicate that it is highly sensitive to DEX exposure (as has previously been reported [22, 23]), and in a sex-specific manner. At the end of the 60-h DEX treatment, *SLC2A1* expression was reduced within the labyrinth of the male and female placenta, a reduction that persisted to near term in the male placenta, but not in the female placenta, where expression was not different from that in the control group. In contrast, in the junctional zone, *SCL2A1* expression was

increased in the DEX-exposed male placenta and decreased in the female placenta at d23. At the end of pregnancy, expression of *SCL2A1* was similar between groups in males, and significantly increased in the female after DEX exposure. These observations illustrate that the placental response to a synthetic stress hormone differs depending on placental region (labyrinth vs. junctional zone), time after exposure (immediately vs. 2 wk later), and fetal sex. As far as glucose transport into and across the placenta is concerned, functional studies are needed to quantify it and to determine the impact of this treatment on fetal development.

IGF1 and *IGFIR* play a key role in promoting fetal and placental growth [39]. DEX significantly reduced the expression of *IGF1* and its receptor, *IGFIR*, in both the male and female placenta, and this decrease persisted until near term. Despite these marked mRNA changes in expression of genes known to affect growth (and of *SLC2A1*, a gene known to affect glucose uptake to the placenta and fetus), we did not observe changes in fetal weight in our study. However, the fetal:placental weight ratio was significantly reduced by the DEX treatment in both males and females. A decreased fetal:placental weight ratio has been shown by others to be associated with compromised fetal and postnatal development [40, 41]. Consistent with this, others and we have shown that the development of the kidney is particularly vulnerable to DEX exposure, with a significant reduction in nephron number seen in the adult offspring of DEX-exposed mothers, but who were of normal weight at birth [15, 42, 43]. This highlights that, while suboptimal conditions in utero may not affect the growth of the fetus, development of fetal organs can still be affected, and the changes in placental gene expression identified here may have played a role.

We observed structural changes in the placenta after DEX exposure. Some changes, specifically the amounts of labyrinth and maternal blood sinusoid, and the ratio of MBS to trophoblast ratio, were affected differently depending on the sex of the fetus (Fig. 2). Precisely how these structural changes are brought about by the brief increase of a glucocorticoid, such as DEX, is unclear. Changes in the expression of *MAP2K1*, *GCM1*, and the *BAX:BCL2* ratio in the placenta suggest that trophoblast within the placenta undergoes considerable remodeling, and that this is sex dependent. These particular genes are known to be intricately involved in trophoblast proliferation [44], structural organization of chorionic villi [45, 46], and apoptotic regulation [47, 48], key processes that would be expected to be involved in the structural changes observed following the DEX treatment. It is important to recognize that significant structural reorganization of the spiny mouse placenta is occurring at the time that the DEX was administered (e.g., the labyrinth is only identifiable after ~20 days gestation), and this is associated with up-regulation of *GCM1* and *MAP2K1* gene expression at this time of pregnancy [49].

An immediate effect of DEX exposure was a decrease in the labyrinth:junctional zone ratio in both male and female placentas, and within the junctional zone there was a marked increase in the amount of trophoblast. This altered labyrinth:junctional zone ratio has been associated with increased metabolic demands of the placenta [50], which may result in placental consumption of a greater fraction of the nutrients that would otherwise pass to the fetus. Near term, the increase in maternal blood sinusoids within the junctional zone of the male placenta, and the opposite effect seen in the female placenta, suggest that placental hemodynamics are altered in quite different ways for the male and female fetus. The architecture of the rodent placenta is such that maternal blood enters the

labyrinth region via large central arteries [51]. The blood then percolates through the labyrinth region, allowing exchange with the fetal circulation, and then enters the maternal venous sinusoids in the junctional zone to return blood to the maternal circulation [51]. Therefore, the changes in tissue mass and maternal blood sinusoid observed in this study might be expected to alter the 'dwell' time of maternal blood within the entire placenta, thereby altering the time for exchange of substances between the maternal and fetal circulations.

The mechanisms by which these sex-specific differences have arisen are not known. There is evidence that males and females differ in their glucocorticoid-specific responses [52–54], and we speculate, on the basis of our findings, that these differences also exist in placental tissues of each sex. It is interesting to consider that the changes we observed in response to DEX were similar between the sexes at d23, a time preceding the functional development of the gonads of both sexes, but that, at d37, when the spiny mouse fetus is well developed, we observed sex-specific responses to this treatment. We therefore hypothesize that fetal sex hormones may have a role in the placental changes observed, although we currently have no direct evidence for this. A limitation in defining these mechanisms is the inability to differentiate whether these effects on the placenta are caused by the glucocorticoids acting directly on the placenta, or result from glucocorticoid-mediated changes in the fetus and/or mother. Future research will investigate both steroid hormone effects and the mechanistic pathways involved.

Overall, this study provides further evidence that male and female placentas are fundamentally different, and that they respond differently to physiological and hormonal challenges that may arise during pregnancy. It is perhaps inappropriate to analyze the rodent placenta as a single entity, as we have shown that the different regions of the placenta respond differently to a stimulus such as excess maternal glucocorticoids given at midgestation. Moreover, the immediate effects of such treatment may either persist, or evolve differently, depending on fetal sex. It is important to note that the sex-specific differences were observed between males and females from the same litter, suggesting that they arise within the placenta itself, and not from an effect of the DEX treatment on the mother. These sex-dependent changes in placental gene expression and structure support the suggestion by Edwards et al. that placental as well as fetal origins should be investigated to explain sex differences in morbidity and mortality that are observed clinically in both fetal and adult life. [55].

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